21 Mark for Review

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy-plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $\left(0, -\sqrt{358}\right)$. If $h(m) = 0$, what is the greatest possible value of m?

Answer Preview:

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21 Mark for Review

The function f is defined by $f(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?



Answer Preview:

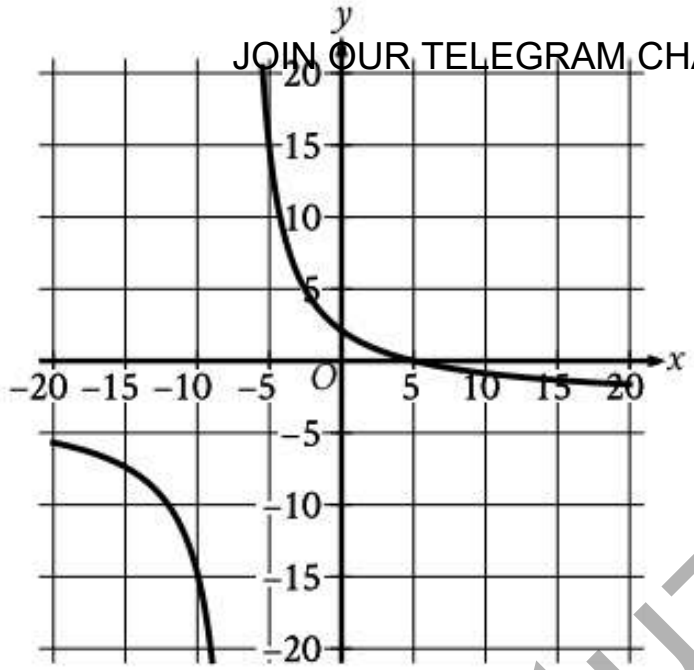
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18 Mark for Review

The function f is defined by $f(x) = \frac{x^2+ax+b}{2x+c}$, where a, b, and c are constants. The graph of the function f in the xy-plane, where $y = f(x)$, does not intersect the line $x = 4$. If $f(5) = f(6) = 0$, what is the value of $a + b + c$?

Answer Preview: 11

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The function f is defined by $f(x) = \frac{ax+b}{x+c} + k$, where a, b, c, and k are constants. The graph of $y = f(x)$ is shown. For how many values of x does $f(x) = 0$?

A.

Three

B.

Two

C.

One

D.

Zero

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The function v is defined by the equation $v(x) = \frac{x^2+bx+c}{(x+5)(x-19)}$, where b and c are constants. In the xy-plane, the graph of $y = v(x)$ passes through the points $\left(0, \frac{66}{95}\right)$ and $(11, 0)$. If $v(q) = 0$, which of the following could be the value of q?

A. -6

B. -5

C. 5

D. 6

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The function j is defined by $j(x) = \frac{10x+5}{3p}$, and the function k is defined by $k(x) = \frac{j(x) \cdot p}{x+1}$, where p is a positive constant. What is the value of $k(9)$?

A. $\frac{19}{6}$

B. $\frac{10}{3}$

C. $\frac{95}{13}$

D. $\frac{85}{3}$

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6 Mark for Review ABC

The function j is defined by $j(x) = \frac{4x+2}{9p}$, and the function k is defined by $k(x) = \frac{j(x) \cdot p}{x+1}$, where p is a positive constant. What is the value of $k(3)$?

(A) $\frac{7}{18}$

(B) $\frac{4}{9}$

(C) $\frac{14}{13}$

(D) $\frac{10}{9}$

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4 Mark for Review

$$f(x) = \sqrt{x^3 + ax^2 + bx + c}$$

The function f is defined by the given equation, where a, b, and c are integer constants. In the xy-plane, the graph of $y = f(x)$ passes through the point $(9, 0)$, and $f(0) = 27$. What is the value of $9a + b$?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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21 Mark for Review

The function f is defined by $f(x) = 11x^3$. The graph of $y = f(-x) + c$ in the xy-plane, where c is a positive integer constant, has an x-intercept at $(r, 0)$ and a y-intercept at $(0, t)$, where r and t are constants. Which of the following must be true about r and t?

A $r < 0$ and $t < 0$

B $r < 0$ and $t > 0$

C $r > 0$ and $t > 0$

D $r > 0$ and $t < 0$

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26 Mark for Review

The function w is defined by $w(r) = \frac{1}{r-8} - \frac{r-5}{-r+4.25}$. What is the greatest possible value of r such that $w(r) = 0$?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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